

ZAVKHAN, Mongolia

WMO: 442160

Lat: 48.8239N

Lon: 93.0983E

Elev: 3081

StdP: 13.13

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.0	-34.2	-42.5	0.5	-37.7	-38.9	0.7	-34.2	21.9	2.9	17.9	-3.9	4.0	320	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.7	61.3	86.5	60.4	83.8	59.5	63.9	84.2	62.3	81.6	61.1	79.7	8.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	76.9	71.3	54.7	71.3	70.4	52.8	66.4	71.0	30.9	85.0	29.6	82.1	28.7	80.2	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	22.7	19.7	DB	-38.0	95.7	8.3	3.1	-44.0	97.9	-48.8	99.7	-53.5	101.4	-59.5	103.7
			WB	-38.8	65.6	8.1	2.5	-44.6	67.4	-49.4	68.9	-54.0	70.3	-59.9	72.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	-16.2	-7.5	22.0	44.9	56.3	68.7	72.8	68.2	55.4	38.9	15.8	-5.5	(6)
(7)		DBStd	32.02	10.26	13.38	13.90	9.26	8.42	6.34	5.15	6.12	8.09	8.53	10.10	10.65	(7)
(8)		HDD50	7902	2051	1609	870	199	38	0	0	39	352	1025	1719		(8)
(9)		HDD65	11620	2516	2029	1335	602	287	36	6	40	302	808	1475	2184	(9)
(10)		CDD50	2322	0	0	1	47	233	562	707	564	200	8	0	0	(10)
(11)		CDD65	565	0	0	0	0	17	148	248	139	13	0	0	0	(11)
(12)		CDH74	5248	0	0	0	10	254	1395	2253	1174	162	0	0	0	(12)
(13)		CDH80	1605	0	0	0	0	50	413	778	336	28	0	0	0	(13)
(14)	Wind	WSAvg	7.2	3.7	4.4	7.1	9.4	10.0	8.7	8.7	8.3	8.0	7.3	6.0	4.2	(14)
(15)	Precipitation	PrecAvg	7.3	0.2	0.2	0.2	0.4	0.6	1.1	1.8	1.2	0.6	0.4	0.4	0.3	(15)
(16)		PrecMax	11.6	0.4	0.3	0.6	0.9	1.0	2.2	4.8	2.6	1.4	0.9	0.9	0.6	(16)
(17)		PrecMin	4.2	0.0	0.1	0.0	0.1	0.2	0.4	0.4	0.3	0.2	0.1	0.2	0.1	(17)
(18)		PrecStd	1.8	0.1	0.0	0.1	0.2	0.2	0.4	0.9	0.6	0.3	0.2	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	29.7	58.4	75.3	85.5	92.2	95.2	91.3	83.2	66.2	44.0	24.1	(19)
(20)		0.4%	MCWB	9.9	22.8	41.6	53.2	57.5	62.8	64.2	61.0	57.3	47.7	35.3	18.9	(20)
(21)		2%	DB	6.8	22.2	51.6	70.1	80.4	88.0	90.9	86.9	79.0	60.7	38.1	16.3	(21)
(22)		2%	MCWB	5.1	17.2	37.7	49.5	55.1	59.9	62.6	61.0	55.6	44.9	30.7	13.8	(22)
(23)		5%	DB	2.1	16.7	46.6	66.1	76.6	84.9	87.3	83.6	74.4	56.6	33.8	12.0	(23)
(24)		5%	MCWB	0.7	13.2	35.0	47.2	53.0	58.6	61.6	59.7	54.1	42.4	28.0	10.2	(24)
(25)		10%	DB	-1.8	11.3	41.6	61.5	72.6	81.9	84.4	80.5	69.9	53.0	29.6	8.5	(25)
(26)		10%	MCWB	-2.7	8.9	32.1	44.6	51.1	58.1	60.6	58.9	52.3	40.5	24.7	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	22.6	42.8	54.1	59.3	67.1	66.9	64.0	60.5	48.5	35.8	19.5	(27)
(28)		0.4%	MCD B	12.7	29.1	55.2	73.5	81.2	87.5	87.4	84.6	80.8	64.7	43.2	25.0	(28)
(29)		2%	WB	4.9	17.4	38.1	50.1	56.0	62.1	64.8	62.3	56.9	45.3	31.3	13.7	(29)
(30)		2%	MCD B	6.7	22.1	50.2	69.3	77.8	81.6	85.4	81.5	72.8	58.9	37.1	16.1	(30)
(31)		5%	WB	0.8	13.3	35.2	47.6	53.9	60.8	63.2	61.1	54.7	43.2	28.1	10.2	(31)
(32)		5%	MCD B	2.0	16.8	46.5	64.7	75.4	79.7	83.0	79.4	72.7	55.7	33.8	11.8	(32)
(33)		10%	WB	-2.8	8.9	32.0	45.1	51.7	59.4	61.9	59.9	52.9	40.9	24.9	7.1	(33)
(34)		10%	MCD B	-1.9	11.4	40.9	60.3	71.3	78.2	80.8	77.6	69.0	52.4	29.7	8.6	(34)
(35)	Mean Daily Temperature Range		MDBR	15.0	19.2	24.3	24.3	22.7	20.5	19.2	19.7	22.2	22.0	16.4	13.6	(35)
(36)		5% DB	MCD BR	17.9	22.4	28.3	28.6	28.1	24.7	23.7	24.0	27.5	26.2	19.7	14.7	(36)
(37)		5% DB	MCWBR	16.6	18.6	18.1	16.1	14.1	10.2	8.7	9.0	12.4	15.2	14.3	12.9	(37)
(38)		5% WB	MCD BR	17.8	22.3	27.0	27.3	26.0	20.4	20.7	20.9	25.5	25.3	19.4	14.2	(38)
(39)		5% WB	MCWBR	16.5	18.5	17.6	16.1	13.9	9.2	8.8	8.6	11.8	15.3	14.3	12.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.282	0.327	0.368	0.369	0.374	0.372	0.352	0.317	0.298	0.275	0.254	(40)	
(41)		taud	2.302	2.248	2.221	2.223	2.265	2.341	2.381	2.408	2.456	2.433	2.320	2.268	(41)	
(42)		Ebn at Noon	260	276	279	278	282	281	280	281	282	269	247	241	(42)	
(43)		Edh at Noon	22	30	37	41	41	38	36	34	29	25	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	429	767	1289	1829	2218	2271	2174	1895	1538	1008	535	344	(44)	
(45)		RadStd	48	59	89	70	107	78	107	80	48	39	56	50	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (5 neighbors)	N/A	-3.85	-4.79	N/A	-1.62	-2.14	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page